Stories Shape Solutions: Balancing Human-Beaver Relationships to Build Resilient Watersheds

COLORADO MOUNTAIN COLLEGE

Cassadie Mefferd | Sustainability Studies

INTRODUCTION

Across the West, drought is reshaping how people relate to water and land. The North American Beaver (*Castor canadensis*), once widely seen as a pest, is now emerging as a partner in restoration. This study explores the social dynamics shaping opportunities to work with beavers in the Upper Yampa Basin.

"As we start seeing these more intense drought years, and we start seeing these bigger wildfires, I do think that there's an opportunity to encourage private landowners who have the flexibility of their land to let beavers move in." - Educator

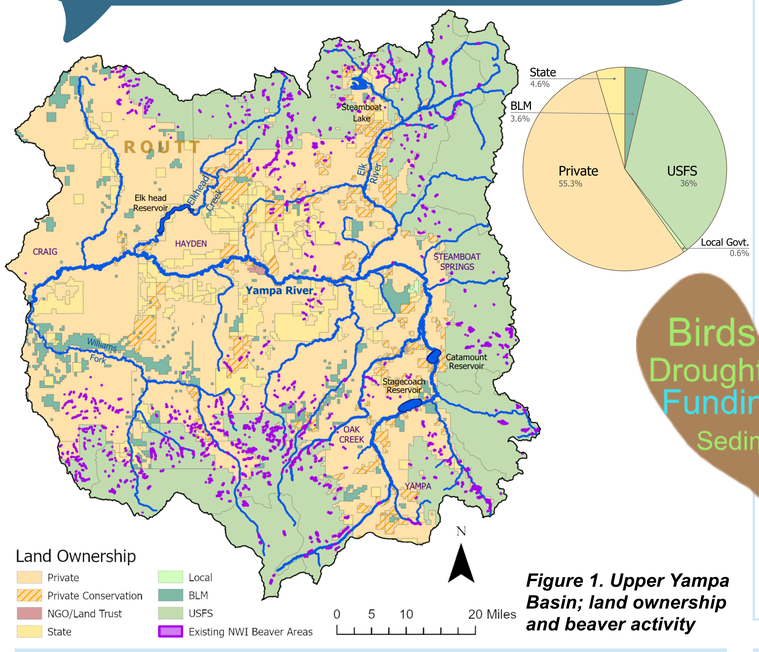


Figure 1. Upper Yampa Basin; land ownership and beaver activity

FINDINGS

"Even though I want to support beaver populations as much as I can and restore habitat for them, I also acknowledge that there is a real balance that has to be struck, and sometimes they just don't belong in places." - Restoration Professional

Benefits (+)	Challenges (-)
Education & outreach opportunity	Mistrust
Nature-based solutions are often more affordable than traditional methods	Difficult to reach rural landowners
Increased rancher recognition drives consumer support	Conflict where beavers interfere with human infrastructure
Wildfire & drought resilience	Flooding of roads or fields
Biodiversity	Landowners report economic losses when beavers interfere with human infrastructure, especially irrigation systems
Slow, store, & spread water	Tree loss

Figure 2. Weighing the Positives & Negatives; colors correspond to (figure 3) & (figure 4)



Figure 3. Word Cloud; frequently mentioned themes

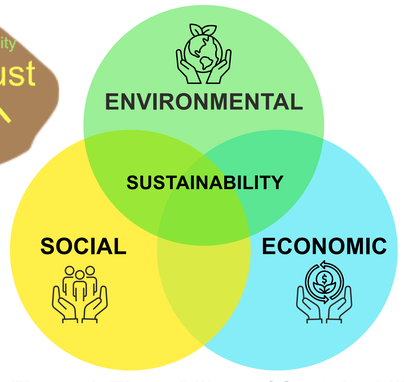


Figure 4. Three Pillars of Sustainability

CONCLUSIONS

- **Restoration requires approaching landowners with respect and offering practical support.**
- **Sustainability depends not only on science, but on the stories, relationships, and realities of people.**
- **Beavers remind us that we are not separate from the systems we manage; we are part of them.**
- **Humans and beavers can flourish when we choose collaboration over control!**

"We don't want to change this land to make it work for us. We have to change ourselves so the land can work with us." - Rancher



Figure 5. Flat Tops Wilderness **Photo Credit: Becky Edmiston**

METHODS

Grounded in transdisciplinarity and mindful inquiry, exploring diverse perspectives.

- **18 semi-structured Interviews:**
  - **Stakeholder Groups: ranchers, restoration professionals, government officials, scientists, and educators.**
- **Community Events:**
  - **Yampa Basin Rendezvous**
  - **Colorado Beaver Gathering**
  - **Yampa Valley Beaver Working Group**
  - **State of the Colorado River Meeting**

DISCUSSION

- **Beavers and humans are part of a complex and interdependent system**
- **Listening to all stakeholder perspectives is essential for effective, long-term success.**
- **Support for beavers is conditional and shaped by personal experience.**
- **Co-existence potential varies by location, depending on ecological and social conditions.**
- **People learn from neighbors and real-world examples, not just information.**
- **This challenges the assumption that increasing education alone will change behavior and improve coexistence.**

NEXT STEPS

- ☑ **Reduce barriers by connecting landowners to affordable, practical coexistence tools.**
- ☐ **Engage rural agriculture communities directly.**
- ☐ **Inspire the next generation of stewards by creating intentional lesson plans and place-based projects for students.**
- ☐ **Apply this research model in other watersheds.**

This Summer...

- ☑ **Interpretive signage at local sites.**
- ☐ **Site tours to increase visibility of successful strategies.**